

AI Recruiter Ranking System

Ranking candidates the way a great recruiter would — not by keywords, but by understanding who fits.

The Problem

Why keyword filters fail



Keyword Blindness

'Python' matches a Data Scientist and a Backend Engineer the same way — context is lost.



Volume Overload

Recruiters review 100s of profiles. Manual review misses the right person every time.



No Depth Signals

GitHub activity, OSS projects, career trajectory — keyword filters ignore all of it.

Our Solution

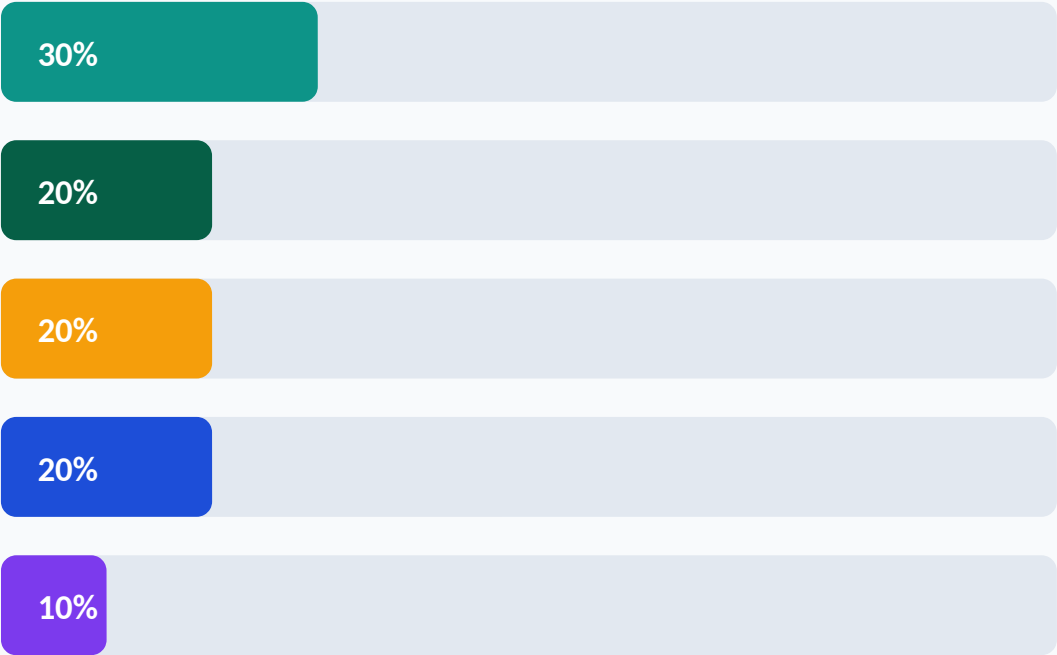
A 5-stage pipeline that thinks like a great recruiter



Every stage is explainable — no black boxes

How We Score

Weighted composite — explainable at every level



30%

20%

20%

20%

10%

Skills Match

Fuzzy matching — avoids Java/JavaScript false positives

Experience Fit

Graduated score around required years of experience

LLM Judgment

Gemini reasons holistically — trajectory, depth, ownership

Semantic Similarity

TF-IDF cosine similarity between JD and profile

Platform Signals

GitHub contributions, OSS projects, certifications

The Key Differentiator

LLM holistic judgment — beyond keywords

What Gemini Evaluates

- ✓ Career trajectory & growth signals
- ✓ Depth vs. breadth of skills
- ✓ Ownership mindset indicators
- ✓ Platform activity (GitHub, OSS)
- ✓ Role-to-role progression logic
- ✓ Seniority level alignment

Sample Output

Candidate: Priya Sharma

Score: 0.847

Rationale:

Strong Python + FastAPI depth with 7 years of backend systems experience. GitHub shows 400+ commits including OSS contributions to a Kafka library. Career shows consistent progression from junior to tech lead.

Strengths: API design, system ownership, OSS
Concerns: Limited Kubernetes experience

Output & Results

Ranked CSV with full explainability — recruiter-ready

Rank	Candidate	Score	Skills	LLM	Rationale (excerpt)
#1	Priya Sharma	0.847	0.89	0.91	Strong FastAPI + Kafka depth, 400+ GitHub commits, tech lead trajectory
#2	Nikhil Reddy	0.821	0.84	0.87	AWS certified, 6y Python, OSS contributor — matches 8/9 must-haves
#3	Aditya Kumar	0.798	0.80	0.85	Senior BE with PostgreSQL+Redis expertise, strong system design background

Full output: 15 ranked candidates with scores, strengths, concerns & rationale

Tech Stack

Clean, modular Python — no prescribed architecture



Gemini 1.5 Flash

Free-tier LLM for JD parsing & holistic candidate evaluation



TF-IDF + sklearn

Semantic similarity retrieval — offline, no model download needed



Fuzzy Skill Matcher

Word-boundary aware — correctly distinguishes Java from JavaScript



CSV Output

15 ranked candidates with composite scores & written rationale



Python 3.12

Modular src/ structure — swap real dataset with zero code changes



11 Tests Passing

Unit tests for scoring logic catch real bugs before submission

Hiring is a judgment call. We built a system that actually makes it.

Not another filter. A recruiter's reasoning — at scale.

[GitHub Repo](#)

[Ranked CSV](#)

[Live Demo](#)